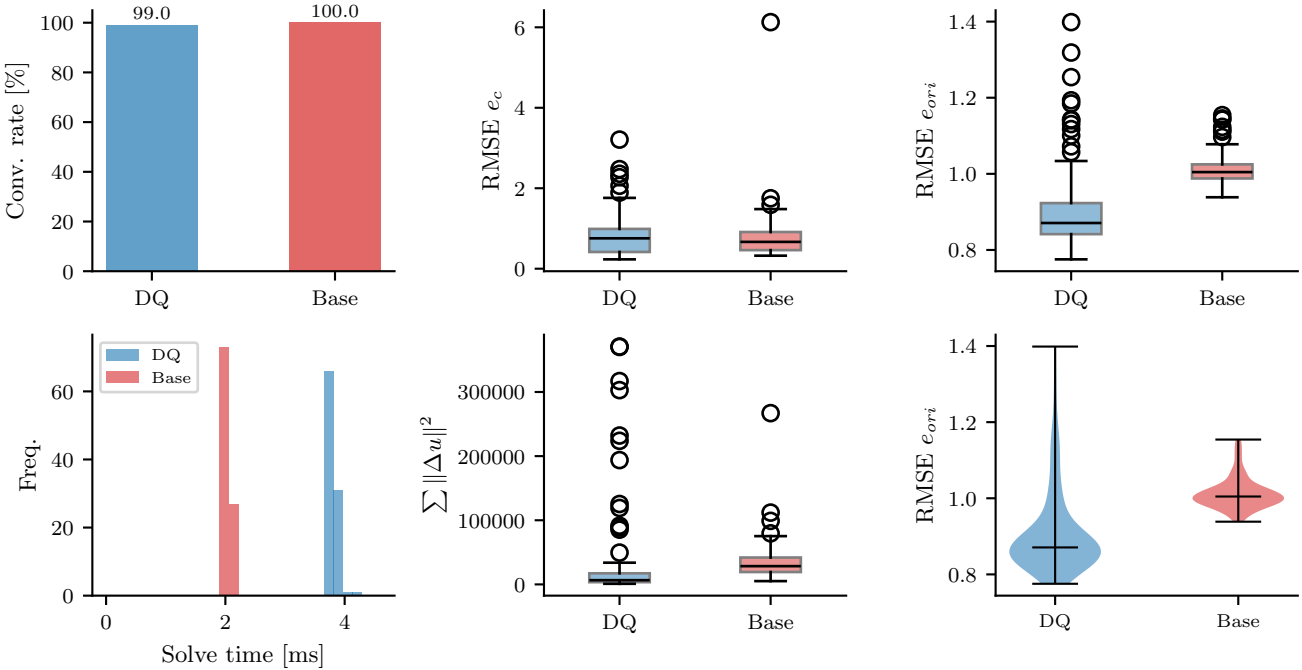Experiment 3: Monte Carlo ($v = 10$ m/s, $N = 100$)